

Module 07

Docker Network and Volume

DevOps end to end services & solutions

Agenda

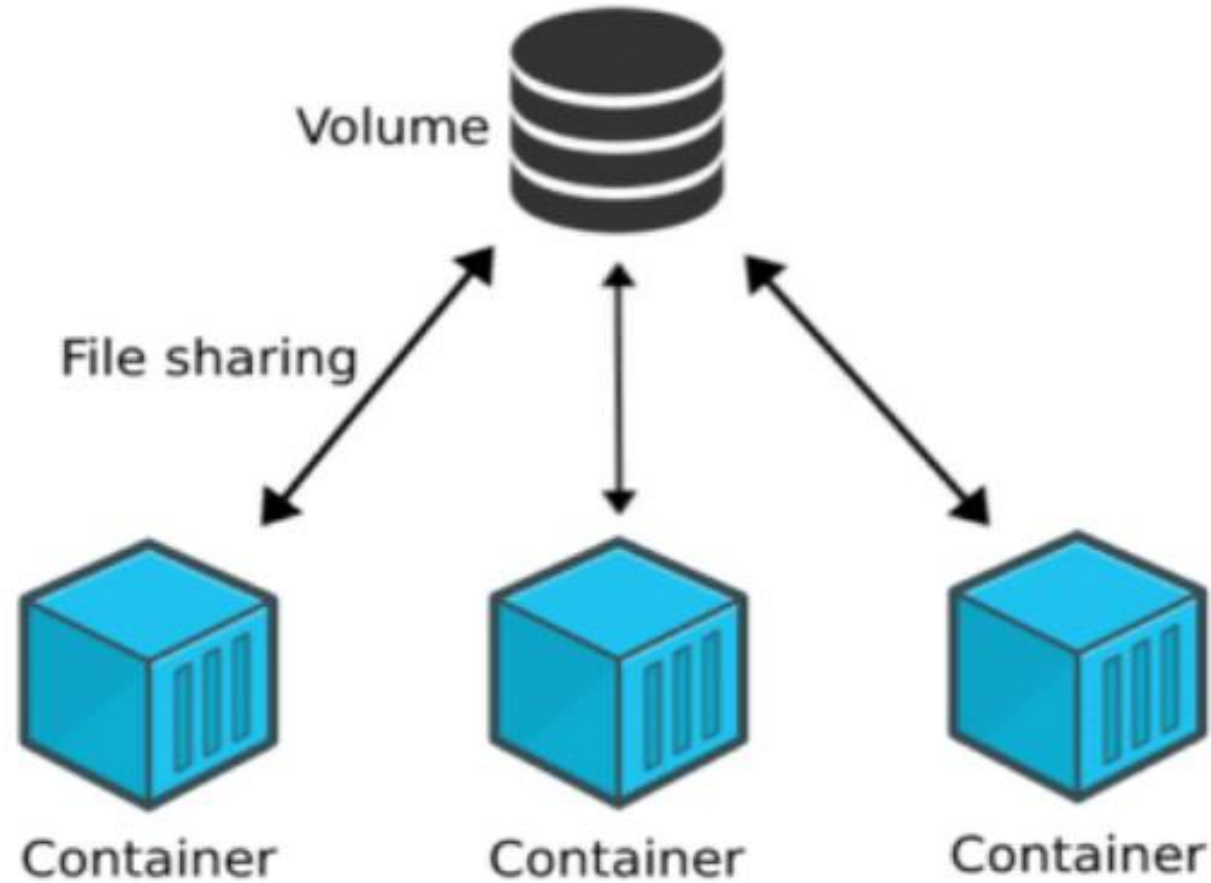
- > Docker Persistent Volume
- > Docker Persistent Volume, example mysql db or vt01
- > Docker Network Bridge
- > Docker Network Host
- > Docker Network Overlay
- > Hands-On – Docker Network

Docker Volume Overview

General

- Docker volume as an external hard drive for your containers.
- By default, any data created inside a container is deleted when the container is removed.
- Volumes allow you to store that data safely on your host machine so it survives restarts, updates, or deletions

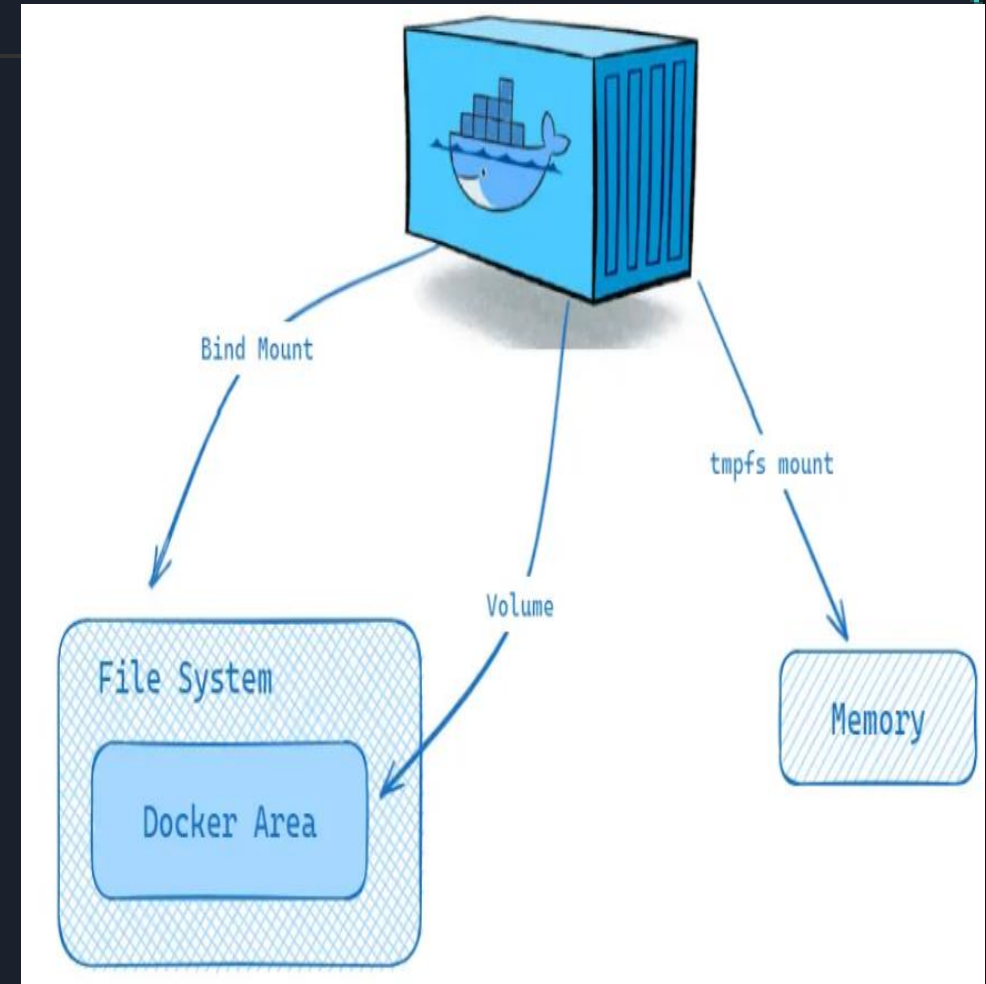
Docker Volume Overview



Docker Volume Overview

Types of Volumes

- > Anonymous Volumes: Docker creates these automatically with a random ID if you don't provide a name. They are harder to manage and reuse.
- > Named Volumes: Created and managed by Docker with a specific name (e.g., my_db_data). These are the most common and recommended.
- > Bind Mounts: These link a specific folder on your computer to a folder in the container. Best for sharing source code during development



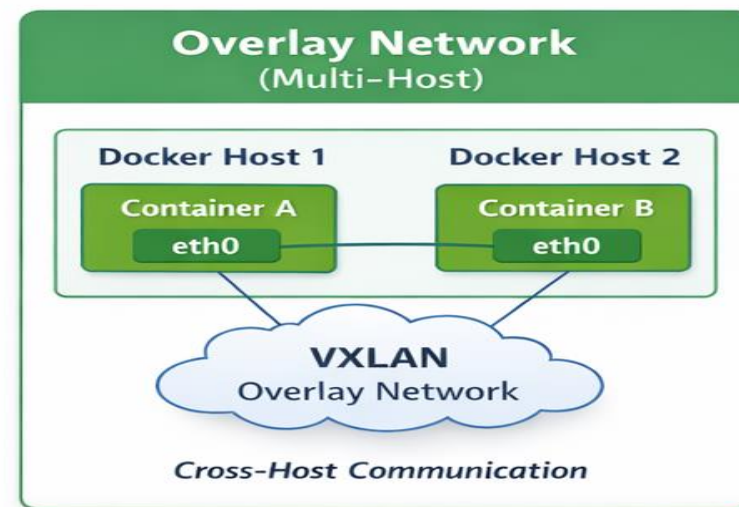
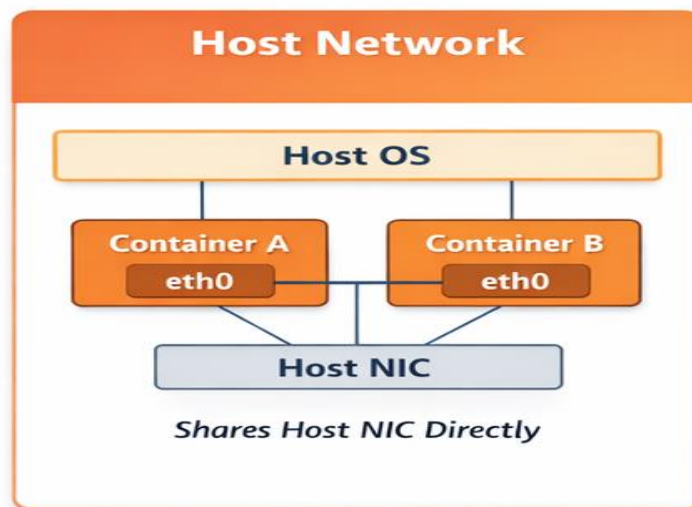
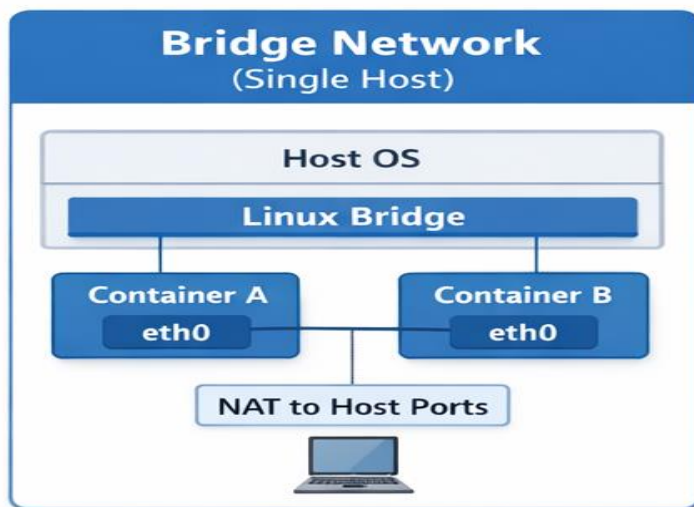
Hands-On – Docker Volumes

- > • Go to Repo – Docker Volumes Lab



Flask

Docker Network General



	Bridge	Host	Overlay
Scope	Single Host	Single Host	Multi-Host
Isolation	✓ Isolated	✗ No Isolation	✓ Cross-Host
Example Use Case	Dev / Test Environments	Monitoring Tools	Swarm / Kubernetes Clusters

Docker Network Bridge network (most common)

Docker network ls

- Uses NAT (iptables)
- Default **virtual LAN**
- Containers get **private IPs**
- Uses **Linux bridge**

Docker network inspect bridge

Isolation

Built-in DNS

Works on single host

Secure by default

- Usage

Single-host apps, microservices

Docker Host Network

Docker network ls

- Container **shares host network namespace**
- No virtual NICs
- No bridge
- No NAT

Docker network inspect bridge

No Isolation

Security risk

- Usage

Monitoring, packet capture, legacy apps

Docker Overlay Network – Multi-host

Docker network ls

- What it is
 - Virtual network **across multiple Docker hosts**
 - Multi-host communication
- Service discovery

Docker network inspect bridge

Secure by default

- Usage

Microservices across multiple nodes

Hands-On – Docker Network

- > • Go to Repo – Docker Network Lab



Flask

Questions?

Thank you for participating.

Feel free to reach out:
<https://dinghy-e2e.com>

